

Experiment No. 1

Experiment Name

Implementation of Fundamental Image Processing Operations using Python and OpenCV

Aim

To develop a comprehensive understanding of fundamental image processing techniques by implementing image acquisition, storage, color space conversion, geometric transformations, and image enhancement operations using Python and OpenCV, thereby establishing the foundation for advanced computer vision applications.

Purpose

Image processing forms the core of modern computer vision systems used in domains such as autonomous driving, healthcare, surveillance, robotics, and industrial automation. Before performing advanced tasks like feature extraction, object detection, or image classification, it is essential to understand how digital images are represented, manipulated, and transformed.

In this experiment, students will implement fundamental image processing operations using Python and OpenCV. They will learn to read, write, convert, resize, and manipulate digital images while understanding how these operations influence image quality and prepare images for subsequent computer vision tasks.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with digital image representation (pixels, resolution, color channels)
- Python libraries: OpenCV (cv2), NumPy, and Matplotlib
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Install and import the required libraries, including OpenCV, NumPy, and Matplotlib.
2. Load a color image from the local system and display it using both OpenCV and Matplotlib.
3. Examine the image properties such as dimensions, resolution, number of channels, and data type.
4. Save the image in different file formats (e.g., JPEG and PNG) and compare the outputs.
5. Convert the image from RGB/BGR to Grayscale, HSV, and LAB color spaces.

6. Perform basic geometric transformations such as resizing, rotation, and horizontal/vertical flipping.
7. Generate the complement (negative) of the image by inverting pixel intensity values.
8. Crop a Region of Interest (ROI) from the image and analyze its characteristics.
9. Display the original and processed images for comparison and observe the effects of each operation.
10. Document the observations and discuss the significance of these preprocessing techniques in computer vision applications.

Questions

1. What is a digital image? Differentiate between grayscale and color images.
2. Explain the difference between RGB, BGR, HSV, and LAB color spaces.
3. What is the purpose of converting an image to grayscale before further processing?
4. Explain the concept of image complement (negative image) and mention its practical applications.
5. Differentiate between image resizing, cropping, and scaling.
6. What is a Region of Interest (ROI), and why is it important in computer vision?
7. Why is OpenCV preferred over conventional image processing libraries for computer vision applications?
8. Explain how image resolution and pixel intensity influence image quality.
9. Mention five real-world applications where basic image preprocessing is an essential step.
10. How do image preprocessing techniques improve the performance of feature extraction and deep learning models?